

LESSON 5-7 • SESSION 1 • PRACTICE • FORM A

Surface Area of Rectangular Prisms

Learning target – I can find the surface area of a rectangular prism by adding the areas of its six faces. • 6.G.A.4

REMEMBER

- **Surface area** is the total area of all the faces, in **square units**.
- A rectangular prism has **3 pairs** of matching faces:
 $SA = 2lw + 2lh + 2wh$.
- A cube has 6 matching faces: $SA = 6 \times (\text{edge} \times \text{edge})$. Open top? Leave that face out.

WORKED EXAMPLE

A closed box is 6 cm $\times$ 4 cm $\times$ 3 cm.**Step 1** Faces: $6 \times 4 = 24$, $6 \times 3 = 18$, $4 \times 3 = 12$.**Step 2** Double each: 48, 36, 24.**Step 3** Add: $48 + 36 + 24 = 108$.**Answer** $SA = 108 \text{ cm}^2$ **WORDS** surface area (*área superficial*) • face (*cara*) • congruent (*congruente*)

PRACTICE

show your work • label units

1. Find the surface area of each solid.

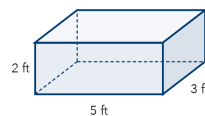
a. cube, edges 8 cm ____ cm^2 b. prism 9 cm $\times$ 6 cm $\times$ 4 cm ____ cm^2 c. prism 10 in $\times$ 5 in $\times$ 3 in ____ in^2 d. cube, edges 6 m ____ m^2

- 2.
- Toy Chest.**
- James stains the whole outside of this chest. One can covers 20
- ft^2
- .

a. Top, front, and end faces _____

b. Surface area _____ ft^2

c. Cans of stain needed _____



Show your work.

- 3.
- Gift Wrap.**
- Kai wraps a box 12 in by 10 in by 7 in. How much paper covers it?

Hint Find three faces, then double each.

Show your work.

Answer _____ in^2

- 4.
- Two Cubes.**
- Cube A has 5 cm edges. Cube B has 7 cm edges. Both are covered in foil.

a. SA of Cube A _____ cm^2 b. SA of Cube B _____ cm^2

Show your work.

- 5.
- Moving Crate.**
- A closed crate is 8 ft by 6 ft by 3 ft. Find its surface area.

Show your work.

Answer _____ ft^2

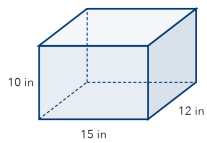
- 6.
- Tessa's Cube.**
- A cube has 12 in edges. Tessa says its surface area is 144
- in^2
- . Is Tessa right?

Show your work.

Answer _____ in^2

- 7. Shipping Carton.** A closed shipping carton is 15 in long, 12 in wide, and 10 in tall. Find its surface area.

Hint Every face has a matching face on the opposite side.



Show your work.

Answer _____ in²

- 8. Painting a Crate.** A crate is 8 ft long, 5 ft wide, and 4 ft tall. One can of paint covers 40 ft².

Hint Round up: part of a can is still a whole can.

- Find the surface area. _____ ft²
- How many cans paint all six faces? _____
- No paint on the bottom: new area and cans _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Jonah's Work.** A box is 10 cm long, 7 cm wide, and 5 cm tall. Jonah wrote: "Surface area = $10 \times 7 \times 5 = 350 \text{ cm}^2$." Explain the mistake and find the surface area.

TEST PRACTICE

choose the best answer

- 10.** A cube with 4 cm edges has a surface area of –
- A** 16 cm² **B** 64 cm² **C** 96 cm² **D** 24 cm²
- 11.** How many pairs of matching faces does a rectangular prism have?
- A** 2 **B** 3 **C** 4 **D** 6

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

548 U	274 Q	1,800 Z	900 C	180 F	144 S	150 R	62 S	184 E	864 A
#2b	#3	#4a	#5	#6	#7	#8a	#8c		

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____